SDC HACKATHON 2026 · DOMAIN 1 – AIVERSECITY FUTURE-READY WORKFORCE

Future Edge

Your career weather forecast for the AI era: which of your skills AI will take over, which will keep you valuable, and the one move to make next.

Team BEBZA · Xiamen University Malaysia
Live: sdc-2026-future-edge.replit.app

SDG 8 Decent Work · SDG 4 Quality Education

THE PROBLEM

Malaysians use AI at work, but nobody is telling them what it will do to their careers

56%
of Malaysian professionals already use AI tools at work

26%
have had any formal AI training

600k+
workers need reskilling in 3–5 years

47→33%
share of tasks done mainly by people, today vs 2030

Today's career tools give a **one-time keyword match** from a static list. AI chatbots can **invent jobs and courses**. Neither tells you *which of your own skills* will still be yours in 3 years.

Hays Asia Salary Guide 2025 and MOSTI (as cited in the Domain 1 brief); WEF Future of Jobs Report 2025, Malaysia profile, p. 169

WHO WE BUILD FOR

One moment: deciding what to learn next, before AI decides for you

⚡ Aiman, final-year student

Final-year Electrical Engineering student. Wants to work in clean energy but doesn't know which skills will age well.

📊 Mei Ling, 6 years in admin

Her data entry and Excel reports are exactly the work AI is taking over. WEF expects data-entry clerk jobs in Malaysia to fall 28% by 2030.

🤖 Ravi, fresh CS graduate

Python, REST APIs and React on his CV – the work AI coding tools are getting best at. What makes him hard to replace?

Plus a customer-service team lead and a junior accountant – five made-up people with realistic made-up CVs (rules: synthetic data only). And HR / L&D managers who plan a whole team's reskilling.

WEF Future of Jobs Report 2025, Malaysia profile, p. 168

THE SOLUTION

From CV to a 3-year forecast and a plan, in about 3 minutes

1 · Upload CV

PDF, Word, photo or text. Name, email, phone, IC, age and gender removed before matching.

2 · Skills radar

Each skill: growing, holding or at risk – and how much AI can do now vs 2029.

3 · Catch the AI

Spot planted mistakes in an AI draft. Measures judgment, not self-rating.

4 · Ranked roles

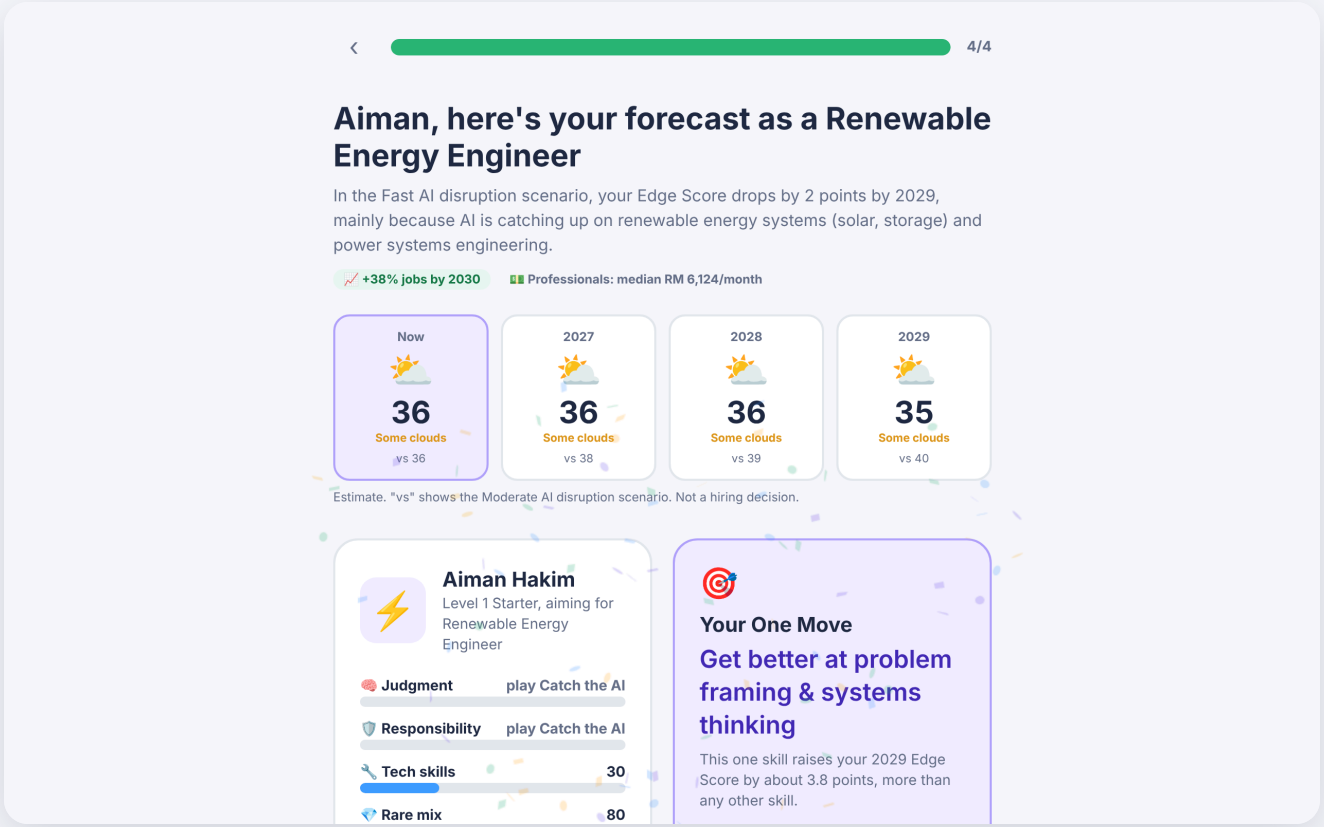
Edge Score 0–100 per year, 2 scenarios, skills gap, real Malaysian pay and demand.

5 · Quest map

Ordered free courses and projects, with hours, SDG tags – starting with your One Move.

Edge Score = how hard you will be to replace by AI, shown as a weather forecast for each year to 2029.

LIVE PROTOTYPE



Aiman's forecast as a Renewable Energy Engineer

- A score for every year, fast vs moderate AI disruption side by side
- Real demand: +38% jobs by 2030 (WEF) and median pay RM 6,124 (DOSM)
- **One Move:** the single skill that raises his 2029 score most
- Always labelled an estimate – never a hiring decision

HOW IT WORKS

Three models, real data, no chatbot guessing

🧠 Model 1 · AI exposure

Text classifier trained on **19,265 human-rated O*NET tasks**. Reads each skill and your own CV sentence.

73.4% accuracy on occupations it never saw (baseline 53.2%)

📈 Model 2 · AI-trend signal

Monthly **arXiv** papers for 10 AI areas, Oct 2023 – Sep 2026. Damped trend with 80% range. Refreshable live.

Backtest error **17.2%** vs 22.7% naive

🕵️ Model 3 · Catch the AI

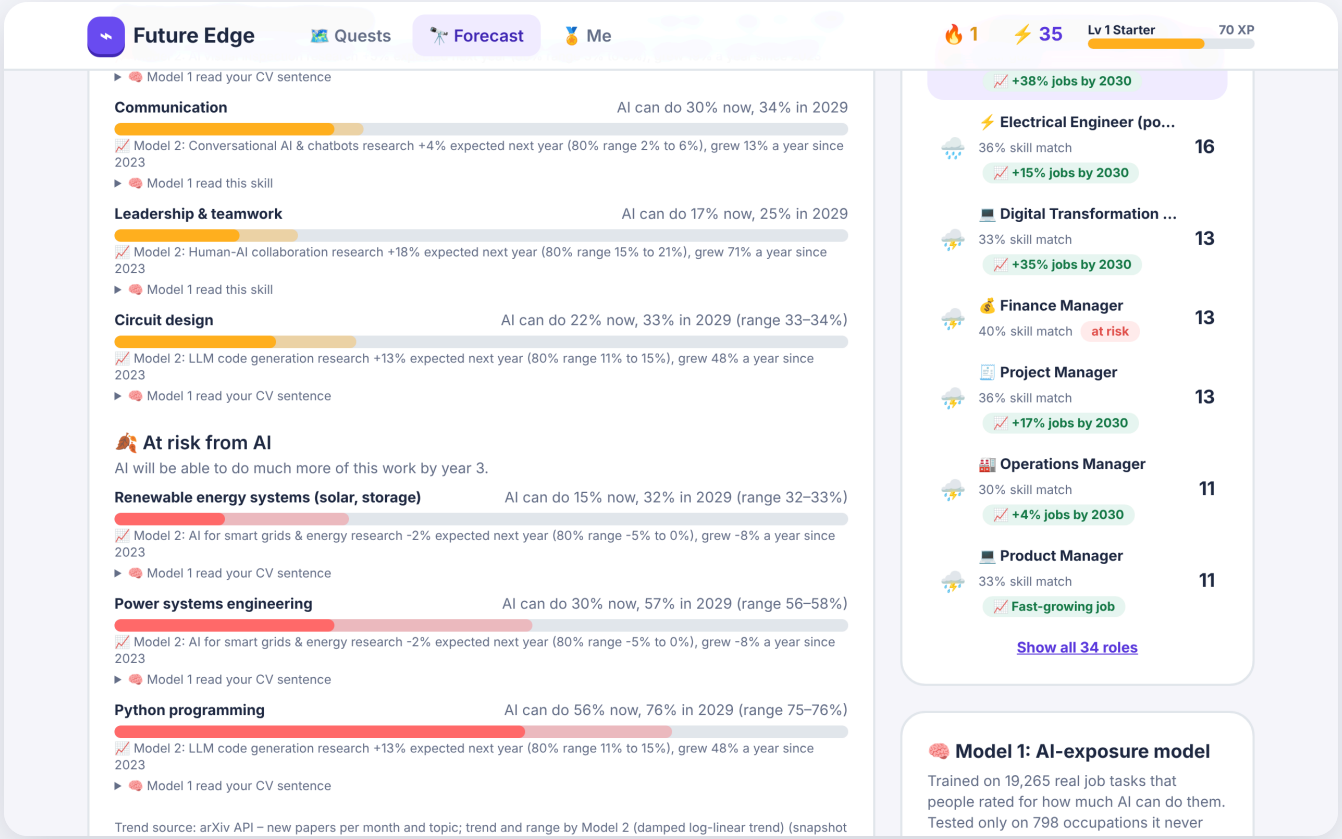
Review an AI-written site note, report or code with **planted mistakes**. Scores judgment and responsibility exactly.

Known answers – no opinions to learn

Fused with **WEF Future of Jobs 2025 (Malaysia)**, **DOSM** wages and **TalentCorp MyCOL** into an explainable Edge Score with a low–high range.

Eloundou et al. (2023) "GPTs are GPTs" labels (MIT); O*NET (CC BY 4.0); arXiv API. Full model cards: docs/MODELS.md

FUTURE SKILLS RADAR · TRANSPARENT BY DEFAULT



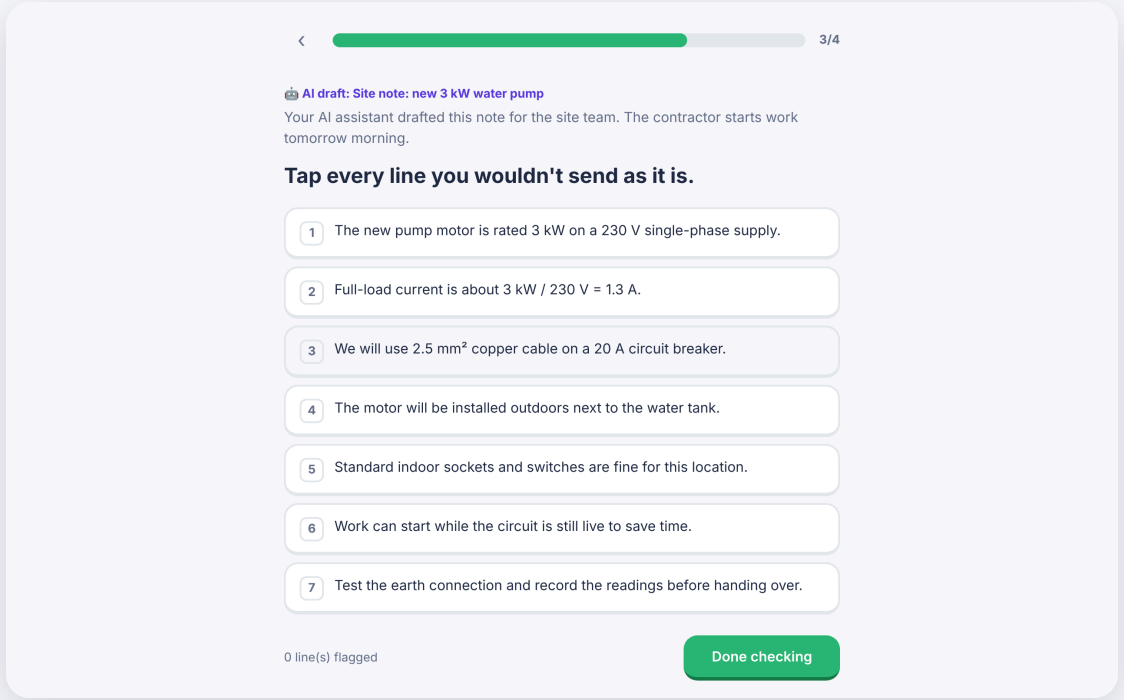
- Every skill shows **its source**: which arXiv trend, which model reading, and the snapshot date
- Change the trend data and **the forecast changes**
- Roles ranked by 2029 Edge Score, flagged **at risk** when AI can do half their work
- Human skills (leadership, communication) gain AI exposure slowly – AI supports them, people own them

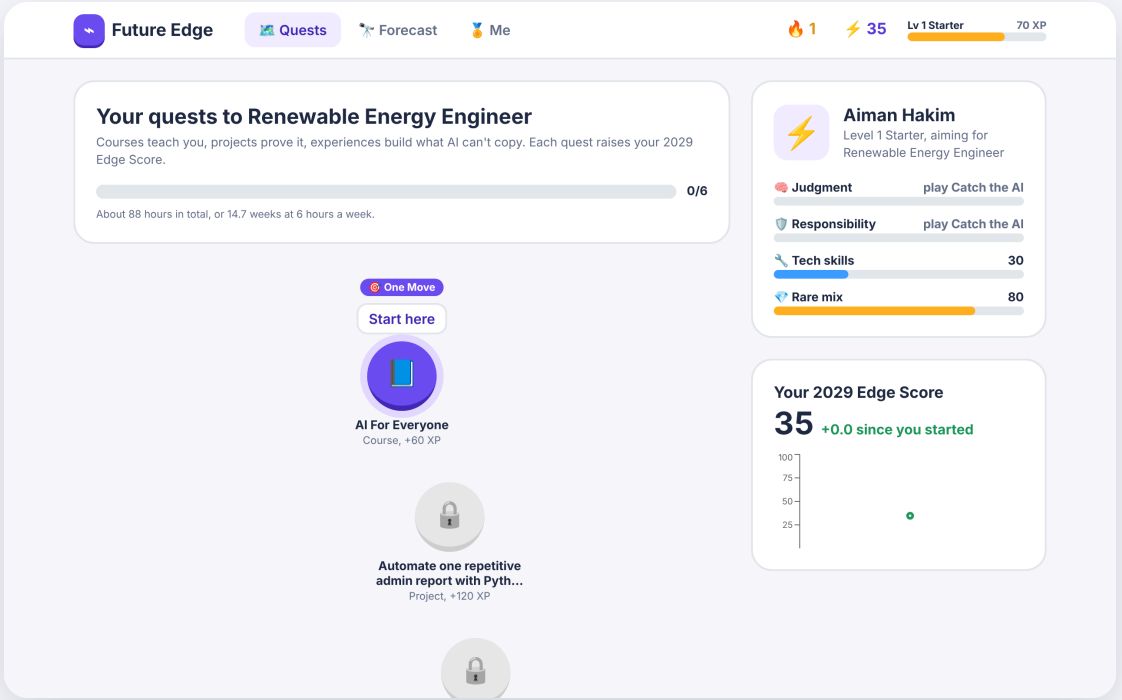
ALL SIX MUST-HAVE FEATURES – WORKING TODAY

MUST-HAVE	IN FUTURE EDGE
1 Skills intake	CV upload (PDF / Word / photo / text) or paste; only catalog skills, with the CV sentence as evidence; user can edit levels
2 AI-trend signal	arXiv monthly counts + Model 2, source and date shown, live refresh button; WEF / DOSM / MyCOL job market
3 Role matching	34 roles ranked by Edge Score with plain-English "why" for every number and a skills gap
4 Reskilling pathway	Ordered courses + projects with hours and weeks; 53 free / low-cost resources, all links checked 24 Oct 2026, HRD Corp option
5 SDG mapping	Every skill, role and step tagged; "your path contributes to..." with reasons; pick what your work should help with and matching roles move up
6 Uncertainty	Moderate vs fast AI scenarios, low–high ranges from Model 2 and Model 3, "estimate, not a hiring decision"

Good-to-haves too: HR team view, EN / BM / 中文 interface, a 12-person fairness check, progress tracking (XP, levels, badges), mobile-friendly – see the next slides.

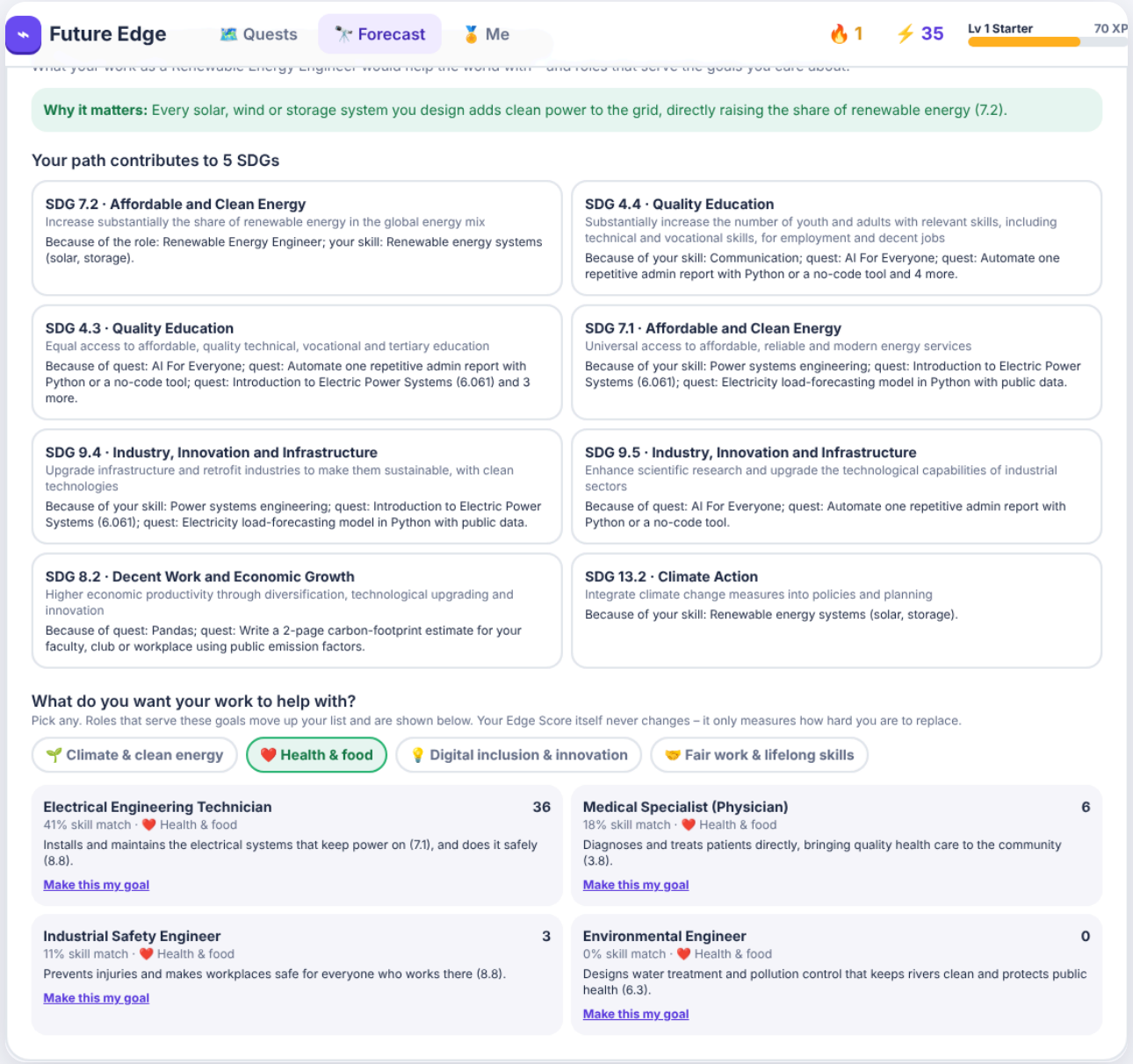
HUMANS STAY IN CHARGE





We measure the skill the AI era rewards – **judging AI output** – and turn the forecast into small quests the user chooses to follow. The app never ranks people for employers.

SDG LINK THAT CHANGES CHOICES

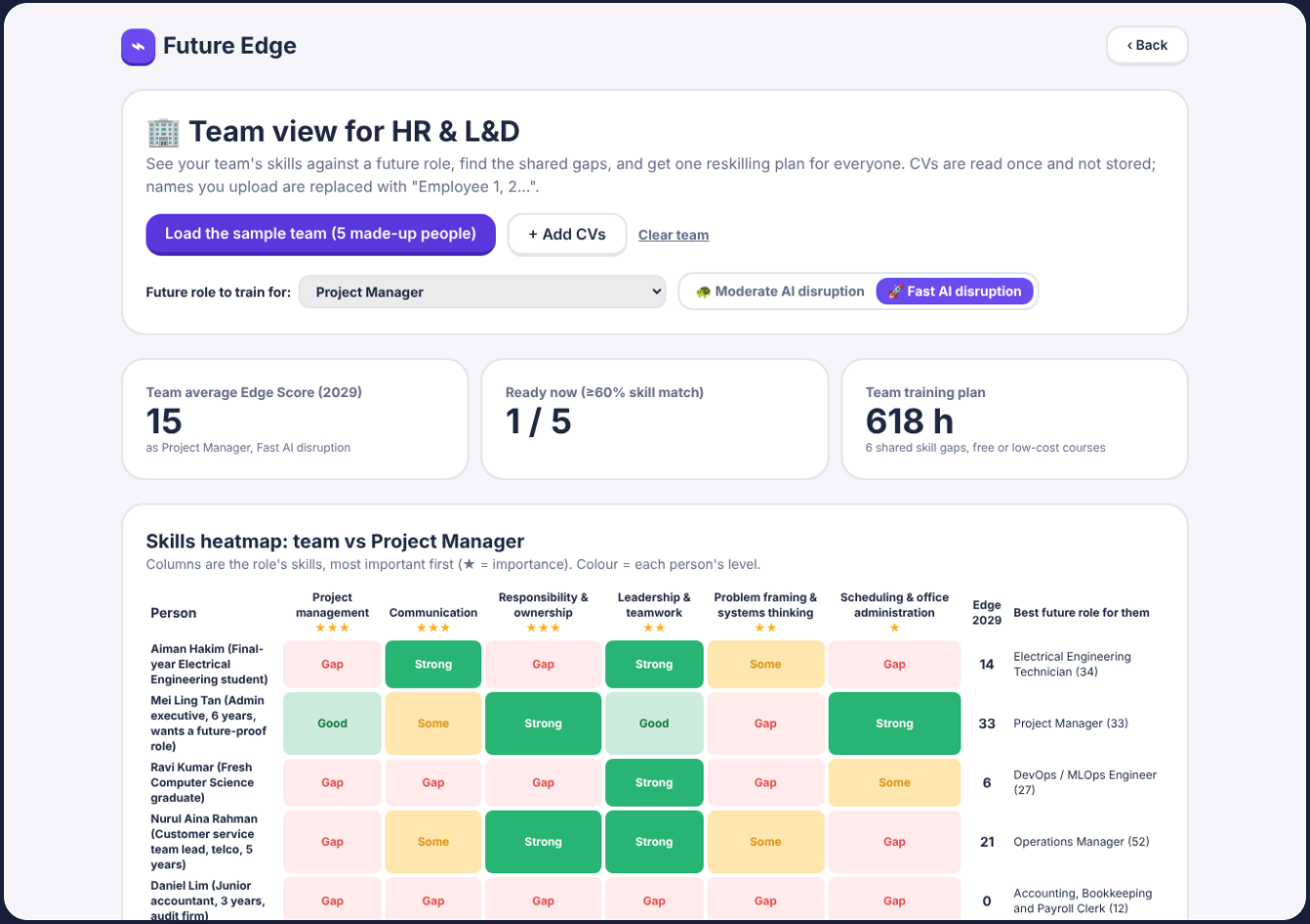


Not decoration: the SDGs shape the plan

- **Why it matters:** one plain line per role on how the work serves its SDG
- **Your path contributes to…** every SDG target, with the reason: the role, your skills, each quest
- **What do you want your work to help with?** Climate, health, digital inclusion or fair work – matching roles are surfaced and lifted
- The Edge Score itself never changes – it stays an honest measure

BEYOND THE MUST-HAVES

Built for teams, fairness and every Malaysian



🏢 HR team view

Skills heatmap of a whole team vs a future role, shared gaps, one training plan with hours, team SDG contribution.

⚖️ Fairness check

Every CV is re-read as 12 invented people – other names, genders, ages, races, religions. The result must be identical.

🌏 EN · BM · 中文

Switch language on every screen. Mobile-friendly, free courses only.

TECHNICAL APPROACH

Simple, explainable, deployed on Replit

Stack

React + Vite + Tailwind → FastAPI (Python) → three in-process models → JSON catalogs. No database, no stored CVs. Optional Gemini API with a JSON schema for photo CVs, every phrase checked against the CV.

Replit (official sponsor)

Imported from GitHub, built and tested in the Replit Shell (55 tests, 53 links checked with open internet), and hosted as a Replit autoscale deployment.

Quality

55 automated tests incl. a fairness test, held-out model evaluations, Apache-2.0 licence, every dataset and AI tool declared. CPU only – no GPU work, so RunPod not needed.

Hardest risk: reliable, updatable trend data. **Fallback:** free arXiv API with a resumable collector; if a refresh fails the app keeps the last good snapshot.

EXPECTED IMPACT · SDG 4.4 AND 8.5

From "AI is coming" to "here is my next move"

≥ 80%

of test users reach ranked roles *and* a sourced plan in ≤ 5 minutes

+40 pts

more users able to name one skill to learn next and a free resource for it (before vs after)

100%

of recommended courses link-checked within 7 days of a demo (53 / 53 today)

Phase 2: user test with 20+ XMUM students and staff · translate course content too · company / region SDG dashboards · job-posting data as a second trend signal.

Baseline: only 26% of Malaysian professionals using AI have formal training (Hays 2025).

TEAM BEBZA · XMUM

AI should be a partner, not a replacement.
Future Edge shows you how to stay the partner.

Azimkhan Bekassyl (lead) · Elnurken Abdigaparov · Bernar Zholdybek · Alibek Berik · Shirvanskiy Zeinal

Try it: **sdc-2026-future-edge.replit.app**
Code: github.com/Magisterofdrinkingcoffee/sdc2026-future-edge